Business model

A testable business proposal, with costs and claims kept separate from measurements.

Question	HeatShift proposal
Who uses it?	Households, students and people planning outdoor activities. The current product supports three airport stations.
Who could pay?	Universities, campus operations, event teams and small logistics businesses may pay for shared dashboards, service support and auditable reports. These are hypotheses; there are no verified customers.
What is the value?	Understand tomorrow's temperature alongside input quality, uncertainty and traceable evidence in one place. Validate whether this improves decisions before claiming impact.
How could it earn?	Keep a basic public educational view. Test an institutional monthly subscription for managed locations, reporting and support. An example PKR 15,000/month price is a research hypothesis, not an offer or validated willingness to pay.
How will it reach people?	Pilot with student/campus groups, interview users and measure repeat use and forecast comprehension. Obtain institution consent before integrating workflows.

Operating costs: prototype versus commercial product

Current prototype uses the existing free static Hugging Face Space and public GitHub repository. Model inference and agents run in the visitor's browser; the hourly job runs on GitHub Actions. No paid hosting or API subscription was purchased. Free tiers are subject to provider terms, capacity, quotas and changes.

At the present hourly design, nominal external volume is one combined METAR request plus three JMA requests per hour: 96 requests/day, before retries and manual dashboard predictions. Full evidence artifacts are retained for 30 days. Monitor upstream limits, action minutes, artifact storage and browser runtime downloads.

Commercial cost item	Planning treatment
Weather API rights / previous-runs access	Obtain the appropriate Open-Meteo commercial plan and quote; amount unverified.
Hosting, scheduling, storage, monitoring	Budget from actual traffic and provider quotes. Static browser compute reduces server inference needs.
Maintenance and customer support	Propose 4-8 staff hours/month initially; this is an estimate, not observed effort. Multiply by actual team hourly cost.
Break-even	Monthly fixed cost / (subscription price - variable cost per institution). Do not claim profitability before measuring both.

Impact and validation plan

Primary outcome: can people correctly explain the forecast, the uncertainty and the data source after using the dashboard? Pilot with a small voluntary group. Measure task completion, time to find a prediction, correct interpretation of the interval, repeat use and usefulness interviews. Compare with the existing way they check weather; record unfavorable feedback too.

Potential wider benefits are learning about data quality and more informed everyday planning. No avoided-loss, carbon-saving, revenue or safety benefit has been measured. Airport estimates do not represent all neighborhood conditions. Keep official weather warnings accessible and make unavailable services visible.

Evidence needed before charging

Recover original training provenance; independently assess performance across seasons; confirm commercial data rights; test mobile usability and startup performance; measure actual support and infrastructure costs; validate payer demand with interviews and a small paid pilot. Define the service scope and support expectations before selling.

Effort, accuracy, speed and memory

Human work hours were not recorded, and substantial integration/documentation work is AI-assisted. Validation MAE is 1.102 C versus 1.235 C persistence, with known January weakness. Warm local Node inference took 80.4 ms for 2,538 rows and used an 11.0 MiB heap snapshot. That excludes startup, networking and total process/browser memory. Quality and insight medians on 5,012 rows were 0.238 s and 0.103 s; traced Python allocations were 2.80 and 1.88 MiB. Browser Python first use can take about a minute.

Sources and pricing caveat

Open-Meteo pricing and license scope; GitHub scheduled workflow behavior. Provider pricing was checked on 21 October 2026. Free-tier access and a running prototype do not establish commercial permission or an uptime guarantee.

Team contributions

Waleed Irfan: developed and improved the prediction model. **Syed Mateen Ali:** developed, managed and maintained both AI agents: the Data Quality Agent and the Insight Agent. **Aman Khan Lodhi:** handled data cleaning, data warehouse management and related data preparation.

Roles supplied by Waleed Irfan. AI assistance is recorded separately in the cumulative AI use log.